

Functional Annotation Report: PP_4379 (UniProt Q88ES4)

β -Ketoacyl-[acyl-carrier-protein] synthase III (FabH / KAS III) of *Pseudomonas putida* KT2440

1. Summary (Answer to the Research Question)

PP_4379 (Q88ES4) encodes the housekeeping β -ketoacyl-[acyl-carrier-protein] synthase III (KAS III, "FabH"; EC 2.3.1.180) of *Pseudomonas putida* KT2440. Its product is a soluble, cytoplasmic condensing enzyme of the bacterial dissociated (type II) fatty-acid synthase (FAS II). Its primary catalytic function is to **initiate** each cycle of de novo fatty-acid biosynthesis by carrying out a decarboxylative (Claisen) condensation of an acyl-CoA primer — physiologically **acetyl-CoA** — with **malonyl-ACP**, producing **acetoacetyl-ACP (β -ketoacyl-ACP)**, **CO₂** and **CoA**. The resulting β -ketoacyl-ACP seeds the reductive elongation cycle that builds straight-chain fatty acids for membrane glycerophospholipids. Catalysis uses the canonical KAS III **Cys–His–Asn** triad (here **Cys114–His237–Asn268**). A notable species-specific feature is that, unlike in *E. coli* where FabH is essential, in *P. putida* FabH is **not essential**: the elongation enzyme FabB provides the in-vivo initiation activity, so PP_4379 is a catalytically competent but functionally redundant initiator.

Annotation caveat (important): The GenBank/UniProt free-text name "*Beta-ketoacyl-acyl-carrier-protein synthase I*" is a **legacy misannotation**. The attached EC number (2.3.1.180), the KEGG orthology (K00648), the eggNOG group (COG0332, FabH), and the Pfam domains (ACP_syn_III, PF08545/PF08541) all unambiguously specify **synthase III (FabH / KAS III)**, *not* synthase I (FabB). This report treats PP_4379 as FabH throughout.

2. Target Verification

Attribute	Value	Source/ Evidence
UniProt	Q88ES4 (unreviewed, TrEMBL)	UniProtKB
Locus	PP_4379; RefSeq WP_003254453.1; GenBank AAN69957	UniProt/ RefSeq
Organism	<i>Pseudomonas putida</i> KT2440 (TaxID 160488)	UniProt
Length / MW	309 aa / 33.2 kDa	UniProt sequence
Domains	ACP_syn_III N (PF08545, res 108–185); ACP_syn_III C (PF08541, res 221–306); Thiolase-like fold (IPR016039); CDD KAS_III (cd00830)	UniProt/ InterPro
EC / KO	EC 2.3.1.180 / K00648 (3-oxoacyl-[ACP] synthase III)	UniProt/ KEGG
Orthology	eggNOG COG0332 (FabH)	UniProt
Pathways	KEGG ppu00061 (Fatty acid biosynthesis); module M00082 (FAS, initiation)	KEGG
Genome position	complement(4,970,636–4,971,565)	KEGG

Verification conclusion: The gene symbol, EC number, orthology, and domain architecture are mutually consistent and match a FabH/KAS III enzyme in the specified organism. Literature on *Pseudomonas putida* FabH exists and applies directly (see §6). This is the correct target; the only discrepancy is the descriptive "synthase I" name, resolved above.

3. Molecular Identity and Structural Evidence

3.1 Domain architecture and fold

PP_4379 has the two-lobed **thiolase-like (condensing-enzyme) fold** characteristic of KAS III, built from the N-terminal (PF08545) and C-terminal (PF08541) ACP_syn_III domains and a five-layer $\alpha\beta\alpha\beta\alpha$ core (Gene3D 3.40.47.10; SUPFAM SSF53901). KAS III enzymes function as **homodimers**, with each active site formed at the dimer interface — the structural basis established by multiple FabH crystal structures (*E. coli*, *S. aureus*, *M. luteus*) [4,5,6].

3.2 Catalytic triad (conserved)

Sequence alignment to *E. coli* FabH places the canonical **Cys–His–Asn** triad at:

Residue (PP_4379)	Sequence motif	<i>E. coli</i> FabH equivalent
Cys114	...SLGCSGY...	Cys112 (nucleophile)
His237	...FCIHQGS...	His244
Asn268	...LETGNTV...	Asn274

The active-site cysteine is the acyl-accepting nucleophile; the His/Asn pair activates malonyl-ACP for decarboxylation and stabilizes the resulting carbanion/oxyanion. This His-Asn-Cys triad is the defining, conserved signature of all characterized FabH enzymes [2], confirming PP_4379 is a catalytically competent condensing enzyme rather than a degenerate homolog.

3.3 Discrimination from the second KT2440 KAS III paralog

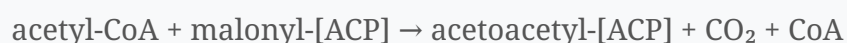
KT2440 contains **two** ~305–309-aa KAS III–domain proteins:

- **PP_4379 (Q88ES4)** — the canonical FabH; **27.6 %** identical to *E. coli* FabH (global Needleman–Wunsch); the **sole** gene assigned KO K00648 and the sole member of the FAS-initiation module.
- **PP_3787 (Q88GD7)** — a **divergent** KAS III (only **14.8 %** identical to *E. coli* FabH and **16 %** to PP_4379), located next to an aminotransferase and a **non-ribosomal peptide synthetase** (PP_3788), consistent with a specialized/secondary-metabolite role rather than housekeeping FAS.

Because PP_4379 is markedly more *E. coli*-FabH-like and is the KEGG-designated FAS-initiation enzyme, it corresponds to the **canonical "FabH1"** of *P. putida* (see §6). (A third, larger KAS-III-domain protein, PP_4545/EC 2.3.1.207, is a distinct enzyme and not a FabH.)

4. Primary Function — Reaction and Substrate Specificity

Reaction (initiation of FAS II):



Mechanism (ping-pong): 1. **Acyl transfer / priming:** the acyl group of the CoA primer (acetyl-CoA) is transferred onto the active-site **Cys114** thiol, releasing CoA and forming an acetyl-enzyme thioester. 2. **Decarboxylative condensation:** **His237/Asn268** promote decarboxylation of malonyl-ACP to a C2 carbanion, which attacks the acetyl-enzyme thioester, yielding **β-ketoacyl(acetoacetyl)-ACP** + CO₂ and regenerating the free cysteine.

Substrate specificity (inferred from close FabH homologs): Canonical FabH enzymes are specific for **short, straight-chain acyl-CoA primers**. For *E. coli* FabH: $K_m(\text{acetyl-CoA}) \approx 40 \mu\text{M}$, $K_m(\text{malonyl-ACP}) \approx 5 \mu\text{M}$; primer preference **acetyl-CoA $\approx$ propionyl-CoA $\gg$ butyryl-CoA**, with **no** activity on hexanoyl-/octanoyl-CoA [1]. For *S. pneumoniae* FabH: $K_m(\text{acetyl-CoA}) = 40.3 \mu\text{M}$, $K_m(\text{malonyl-ACP}) = 18.6 \mu\text{M}$, again preferring straight short-chain CoA primers [2]. Given PP_4379's canonical primer-binding architecture and highest similarity to *E. coli* FabH among KT2440 paralogs, its **physiological primer is acetyl-CoA**, committing acetate/acetyl units into straight-chain fatty acids. (Chain-length/branched-substrate selectivity in FabH is governed by the size of the primer pocket and residues such as the L9-loop/Phe near the triad — the structural determinant that distinguishes acetyl-selective from bulkier-primer FabH enzymes [5,6].)

Downstream fate of the product: acetoacetyl-ACP enters the elongation cycle — reduction (FabG), dehydration (FabA/FabZ), reduction (FabI) — and is elongated by the KAS I/II condensing enzymes FabB (PP_4175) and FabF (PP_1916/PP_3303) to yield C16–C18 acyl-ACPs used for membrane phospholipid synthesis.

5. Localization

PP_4379 is a **soluble cytoplasmic** enzyme. It is a component of the bacterial **dissociated (type II) fatty-acid synthase**, whose enzymes are individual cytosolic proteins acting on substrates tethered to the small acidic **acyl carrier protein (ACP, PP_1915)** [1]. The 309-aa sequence contains no signal peptide and no transmembrane segment. Its products (long acyl-ACPs) are handed to the membrane-associated acyltransferases (PlsX/PlsB/PlsC) at the cytoplasmic face of the inner membrane, but the FabH condensation reaction itself occurs in the cytoplasm.

6. Biological Process, Pathway Context, and Regulation

6.1 Pathway role

PP_4379 catalyzes the **committed initiation step** of **de novo fatty-acid biosynthesis** (FAS II; KEGG module M00082), the pathway that supplies acyl chains for **membrane glycerophospholipids** and, in *P. putida*, ultimately for storage polymers such as medium-chain-length polyhydroxyalkanoates (via β -oxidation/FAS intermediates).

6.2 Non-essentiality and functional redundancy in *Pseudomonas* (key nuance)

Unlike the *E. coli* paradigm — where FabH is the essential, sole initiator — the *Pseudomonas* FAS-initiation architecture is different:

- **Guo et al. (2024)** showed that *P. putida* F1 carries **two functional fabH genes** (PpfabH1 and PpfabH2) but **neither is essential**; **PpFabH1 is a canonical KAS III similar to *E. coli* FabH**, whereas PpFabH2 condenses malonyl-ACP with short/medium-chain acyl-CoAs. The genuine in-vivo initiator is **FabB** (the KAS I elongation enzyme), which decarboxylates malonyl-ACP and condenses the acetyl-ACP product with malonyl-ACP [3]. PP_4379's direct F1 ortholog is **Pput_1488** (79 % identity; conserved genomic synteny), and PP_4379 is the KT2440 counterpart of the **canonical PpFabH1**.
- In the related species *P. aeruginosa*, **deletion of all four KAS III genes yields wild-type growth**, and FAS is instead initiated by the divergent KAS I/II enzyme **FabY (PA5174)** [7].

Interpretation: PP_4379 is a fully competent acetyl-CoA-specific FabH that *can* prime FAS, but in *P. putida* the indispensable initiation flux is provided redundantly by FabB. PP_4379 therefore contributes to and can modulate the pool of short-chain β -ketoacyl-ACP primers rather than being an indispensable gatekeeper.

6.3 Regulation

FabH is a physiological control point of FAS: its activity is **feedback-inhibited by long-chain acyl-ACP end-products** (potency increasing with chain length from C12 to C20; competitive with malonyl-ACP), coupling initiation flux to cellular demand for membrane lipids [1]. This makes PP_4379 a regulatory node even where it is not strictly essential.

6.4 Genomic context (observation)

PP_4379 is not located within the housekeeping *fabD*–*fabG*–*acpP*–*fabF* operon (PP_1913–PP_1916); instead it sits within a **flagellar gene cluster** (flanked by flagellin PP_4378 and hook-associated proteins PP_4380/PP_4381), and its F1 ortholog Pput_1488 is likewise flanked by flagellin/FlaG genes and a nucleotide-sugar epimerase/dehydratase (Pput_1486). This synteny is conserved between strains. While its biochemical assignment as the housekeeping FabH is secure, the consistent co-localization with flagellar/glycosylation-island genes is an intriguing observation that could merit experimental follow-up (e.g., a possible ancillary role supplying acyl primers to a flagellin-glycosylation or lipid-modification pathway). This is **not established** and is flagged only as a hypothesis.

6.5 Pharmacological relevance

FabH/KAS III is a validated antibacterial drug target; structure-based inhibitors bind at/near the catalytic triad, and resistance mutations map to the triable region [4,5]. PP_4379 is the *P. putida* representative of this target class, though its non-essentiality in *Pseudomonas* tempers its value as a lethal target in this genus.

7. Supported and Refuted Hypotheses

Supported - H1 — PP_4379 is a KAS III/FabH (EC 2.3.1.180), not a KAS I: **supported** (domains PF08545/PF08541, KO K00648, COG0332, conserved Cys114–His237–Asn268 triad). - H2 — It initiates FAS II via acetyl-CoA + malonyl-ACP condensation, cytoplasmic: **supported**

(biochemistry of canonical FabH; pathway/module assignment). - H3 — It is the canonical "FabH1" of *P. putida*, more *E. coli*-FabH-like than the paralog PP_3787: **supported** (27.6 % vs 14.8 % identity; sole K00648; orthology to characterized PpFabH1/Pput_1488). - H4 — FabH is dispensable in *P. putida* because FabB initiates FAS: **supported** by direct experimental evidence [3] and consistent with *P. aeruginosa* [7].

Refuted / corrected - The database name " β -ketoacyl-ACP synthase I" (implying FabB/KAS I): **refuted** — all orthogonal evidence indicates KAS III (FabH).

8. Limitations and Future Directions

- **No direct biochemistry on PP_4379 itself.** Function is inferred from unambiguous orthology, conserved catalytic residues, and detailed characterization of close homologs (*E. coli*, *S. pneumoniae*, *P. putida* F1 PpFabH1). Direct enzymology (primer chain-length profile, K_m/k_{cat} , structure) of the KT2440 protein has not been reported.
 - **Substrate specificity is inferred, not measured.** Acetyl-CoA preference is predicted from sequence/architecture; medium-chain activity (as seen for the paralogous PpFabH2) has not been excluded for PP_4379.
 - **Structure is predicted only** (AlphaFold model available; no experimental structure of PP_4379).
 - **Genomic flagellar-cluster context is unexplained.** Whether PP_4379 has an ancillary role linked to flagellar glycosylation is an open, testable question (targeted deletion + lipidomic/glycomic phenotyping).
 - Recommended experiments: recombinant expression + in-vitro condensation assays with an acyl-CoA panel; Δ PP_4379 (and Δ PP_4379 Δ fabB) phenotyping; crystallography/cryo-EM.
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9. References

1. Heath RJ, Rock CO. *Inhibition of β -ketoacyl-ACP synthase III (FabH) by acyl-ACP in E. coli.*

(

P 8631920

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 — reaction, kinetics, primer specificity, acyl-ACP feedback inhibition.

2. Khandekar SS, et al. *Identification, substrate specificity, and inhibition of S. pneumoniae FabH.*
 (P 11375394)
 — conserved His-Asn-Cys triad; kinetics; short straight-chain primer preference.
3. Guo Z, et al. *Diversity in fatty acid elongation enzymes: FabB initiates FAS in P. putida F1.*
 (P 38335573)
 — two non-essential *P. putida* fabH genes; PpFabH1 canonical; FabB is the initiator.
4. McKinney DC, et al. *Antibacterial FabH inhibitors ... resistance mutations near the catalytic triad.*
 (P 27626097)
 — FabH as validated antibacterial target.
5. Qiu X, et al. *Crystal structure and substrate specificity of S. aureus FabH.*
 (P 15987898)
 — fold, primer-pocket determinants of specificity.
6. Pereira JH, et al. *Structure of FabH ... branched fatty acids in M. luteus.*
 (P 22993086)
 — initial condensation step; L9-loop/Phe specificity determinant.
7. Yuan Y, et al. *FAS in P. aeruginosa is initiated by FabY, not FabH.*
 (P 22753059)
 — KAS III dispensable in *Pseudomonas*.

Additional database evidence: UniProt Q88ES4; KEGG ppu:PP_4379 (K00648; ppu00061; M00082); InterPro IPR013747/IPR013751/IPR016039; Pfam PF08545/PF08541; eggNOG COG0332.